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Robust Online Learning over Networks

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Outline

1. Distributed Operator Theoretical (DOT) ADMM
2. Convergence analysis and novelty w.r.t. the state-of-the-art
3. Numerical simulations
4. Extensions and applications

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Problem of interest

We consider a network of n agents linked according to an undirected, connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each agent $i \in \mathcal{V}$ has a vector state $x_i(k) \in \mathbb{R}^p$ with $p \in \mathbb{N}$, and has access to a time-varying local cost $f_{i,k} : \mathbb{R}^p \to \mathbb{R} \cup \{+\infty\}$, $k \in \mathbb{N}$. The objective of the network is to solve the optimization problem $\min_x \sum_{i \in \mathcal{V}} f_{i,k}(x)$, which can be reformulated in the following distributed form:

$$\begin{aligned} &\min_{x_i} \sum_{i \in \mathcal{V}} f_{i,k}(x_i) \\ &\text{s.t.} \quad x_i = x_j \text{ if } (i,j) \in \mathcal{E} \end{aligned} \tag{1}$$

whose set of solutions is denoted by $\mathcal{X}_k^\star$. We consider the following challenging scenario:

- Local datasets available to the agents change over time;
- Asynchronous processing/updates;
- Inexact local computations;
- Unreliable (faulty and noisy) communications.

Assumptions on the challenging scenario

Assumption 1 (local datasets change over time)

At each time $k \in \mathbb{N}$, the local cost functions $f_{i,k}$ are proper, lower semi-continuous, and convex

Assumption 2 (local datasets change over time)

The set of solutions $\mathcal{X}_k^\star$ is non-empty at each time $k \in \mathbb{N}$. In addition, the maximum distance between two solutions at consecutive times is upper-bounded by a nonnegative constant $\sigma \geq 0$.

Assumption 3 (asynchronous updates and faulty communications)

At each time step, each node $j \in \mathcal{V}$ has completed its local computation and successfully transmits data to its neighbors $i \in \mathcal{N}_j$ with probability $p_{ij} \in (0, 1]$.

Assumption 4 (inexact local computations and noisy communications)

Each node $i \in \mathcal{V}$ updates its local variable and receives information from neighbors with an additive noise such that the overall error $e_{ij}(k)$ is bounded in expectation by $\mathbb{E}\left[\|e_{ij}(k)\|\right] \leq \nu_e < \infty$.

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The Distributed Operator Theoretical (DOT) ADMM Protocol

Input: For each agent $i \in \mathcal{V}$ initialize the auxiliary variables $\{z_{ij}(0)\}_{j\in\mathcal{N}_i}$; choose the relaxation $\alpha \in (0,1)$ and the penalty $\rho > 0$.

Output: Each agent returns $x_i(k)$ that is an (approximated) solution to the distributed optimization problem in eq. (1).

for $k = 1, 2, \ldots$ **each active agent** $i \in \mathcal{V}$

// asynchronous computations and inexact updates

receives a local cost $f_{i,k}$ and applies the local update

$$x_i(k) = \text{prox}_{f_{i,k}}^{1/\rho\eta_i}\left(\frac{1}{\rho\eta_i}\sum_{j\in\mathcal{N}_i} z_{ij}(k-1)\right) \tag{2}$$

for each agent $j \in \mathcal{N}_i$

transmits the packet

$$y_{i\to j}(k) = 2\rho x_i(k) - z_{ij}(k-1)$$

end for

for each packet $y_{j\to i}$ **received by agent** $j \in \mathcal{N}_i$

// noisy communications with packet loss

updates the auxiliary variable

$$z_{ij}(k) = (1-\alpha)z_{ij}(k-1) + \alpha y_{j\to i}(k) \tag{3}$$

end for

end for

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Theorem 1 (Linear convergence)

Consider an online distributed optimization satisfying under Assumptions 1-2, and a connected network of agents that solves it by running DOT-ADMM under Assumptions 3-4.
If the DOT-ADMM operator T_k, ruling the updates of the auxiliary variable

$$z(k) = \mathsf{T}_k(z(k-1))$$

is γ-metric subregular, then:

i) there is an upper bound to the distance between the current solution $x(k)$ and the set of optimal solutions $\mathcal{X}_k^\star$ that holds in mean for all $k \in \mathbb{N}$, and this bound has a linearly decaying dependence on the initial condition:

$$\mathbb{E}\left[d_{\mathcal{X}_k^\star}(x(k))\right] = O\left(\mu^k d_{\mathsf{T}_0}(x(0)) + \frac{1-\mu^k}{1-\mu}(\nu_e + \sigma)\right), \qquad \mu \in (0,1). \tag{4}$$

ii) there is an upper bound for $d_{\mathcal{X}_k^\star}(x(k))$ that holds almost surely when $k \to \infty$, and this bound does not depend on the initial condition x(0):

$$\limsup_{k\to+\infty} d_{\mathcal{X}_k^\star}(x(k)) = O\left(\frac{\nu_e + \sigma}{1-\mu}\right). \tag{5}$$

Metric subregularity instead of strong convexity

Definition 1

An operator $\mathsf{T} : \mathbb{R}^n \to \mathbb{R}^n$ is metric subregular if there is a positive constant $\gamma > 0$ such that

$$d_{\mathsf{T}}(x) \leq \gamma \|(\mathsf{Id} - \mathsf{T})x\|, \qquad \forall x \in \mathbb{R}^n. \tag{6}$$

When γ is known, T is said to be γ-metric subregular.

Remarks:

- Metric subregularity of the operator is sufficient for linear convergence;
- Strong convexity of the local cost is neither necessary nor sufficient or linear convergence:
  - Non necessity: due to Theorem 1, examples are linear and logistic regression problems;
  - Non sufficiency: The strongly convex costs $f_i(x_i) = \sqrt[3]{x_i^4} + \frac{\epsilon}{2} x_i^2$ yield a sub-linear convergence

Question: How to verify that the operator is metric subregular?

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Question: How to verify that the operator is metric subregular?

A sufficient condition for metric subregularity

Proposition 1

An operator $\mathsf{T}:\mathbb{R}^n \to \mathbb{R}^n$ is metric subregular if there is a matrix $A \in \mathbb{R}^{n\times n}$ and two offsets $b_{\mathsf{L}}, b_{\mathsf{U}} \in \mathbb{R}^n$ such that T is lower and upper bounded (component-wise) by the affine operators $\mathsf{L}(z) := Az + b_{\mathsf{L}}$ and $\mathsf{U}(z) := Az + b_{\mathsf{U}}$, respectively, yielding

$$\mathsf{L}(z) \leq \mathsf{T}(z) \leq \mathsf{U}(z), \qquad \forall z \in \mathbb{R}^n. \tag{7}$$

Consider *empirical risk minimization* (ERM) problems, in which the local cost of each agent $i \in \mathcal{V}$ is defined over the data set $\{a_{i,h}, b_{i,h}\}_{h=1}^{m_i}$, $m_i \in \mathbb{N}$:

$$f_i(x) = \sum_{h=1}^{m_i} g(x, a_{i,h}, b_{i,h}) \tag{8}$$

Examples:

- Linear regression problems with $g(x, a_{i,h}, b_{i,h}) = \frac{1}{2}(a_{i,h}^\top x - b_{i,h})^2$;
- Logistic regression problems with $g(x, a_{i,h}, b_{i,h}) = \log\left(1 + \exp\left(-b_{i,h} a_{i,h}^\top x\right)\right)$.

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Corollary 1 (Particular cases)

Consider the scenario of Theorem 1 and the following simplified cases:

(a) the cost functions $f_{i,k} = f_i$ are static, i.e., $\sigma = 0$;

(b) communications are noiseless and computations are exact, i.e., there are no additive errors $\nu_e = 0$;

(c) communications are synchronous.

Then the results of Theorem 1 become:

i) (a) implies that the distance $d_{\mathcal{X}_k^\star}(x(k))$ converges linearly to $O(\nu_e)$ in mean;

ii) (b) implies that the distance $d_{\mathcal{X}_k}(x(k))$ converges linearly to $O(\sigma)$ in mean;

iii) $(a) \wedge (b)$ implies that the distance $d_{\mathcal{X}_k}(x(k))$ converges linearly to zero in mean square with rate μ^2; moreover, $x(k)$ almost surely converges to the set of solutions $\mathcal{X}^\star$;

iv) $(a) \wedge (b) \wedge (c)$ implies that $x(k)$ converges linearly with rate μ^2 and almost surely to the set of solutions $\mathcal{X}^\star$.

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Set-up

The considered set-up is a network of $N = 10$ nodes, exchanging information through a random graph topology of 20 edges, that want to solve an online logistic regression problem with local costs

$$f_{i,k}(x) = \sum_{h=1}^{m_i} \log\left(1 + \exp\left(-b_{i,h,k} a_{i,h,k} x\right)\right)$$

where $x \in \mathbb{R}^p$, $p = 16$, is the vector of weights and intercept to be learned, and $\{(a_{i,h,k}, b_{i,h,k}\}_{h=1}^{m_i}\}$ with $a_{i,h,k} \in \mathbb{R}^{1\times p}$, $b_{i,h,k} \in \{-1, 1\}$ are the $m_i = 20$ feature vectors and class pairs at time $k \in \mathbb{N}$.

Inexact local updates

Remark: the proximal of $f_{i,k}$ does not have a closed form! Thus, each node compute an inexact solution via accelerated gradient descent, terminating when the distance between consecutive iterats is smaller than a threshold $\theta > 0$.

Table: Computational time of local updates and asymptotic error for a static logistic regression problem.

Threshold	Comp. time [s]	Asymptotic err.
$\theta = 10^{-14}$	3.47×10^{-3}	4.14×10^{-14}
$\theta = 10^{-12}$	2.84×10^{-3}	3.65×10^{-12}
$\theta = 10^{-10}$	2.42×10^{-3}	4.88×10^{-10}
$\theta = 10^{-8}$	1.95×10^{-3}	5.30×10^{-8}
$\theta = 10^{-6}$	1.39×10^{-3}	1.01×10^{-5}
$\theta = 10^{-4}$	8.88×10^{-4}	5.73×10^{-4}
$\theta = 10^{-2}$	4.12×10^{-4}	9.71×10^{-2}

Quantized communications

Assume that each agent i can only send the quantized version $q(x)$ of a message $x \in \mathbb{R}^p$, namely

$$[q(x)]_j = \begin{cases} \underline{q} & \text{if } [x]_j < \underline{q} \\ \delta\lfloor [x]_j/\delta \rfloor & \text{if } \underline{q} \le [x]_j \le \bar{q}\,, \\ \bar{q} & \text{if } [x]_j > \bar{q} \end{cases} \quad j \in \{1, \dots, p\}$$

with $\bar{q} = -\underline{q} = 10$, and $\delta > 0$ the quantization level.

Table: Asymptotic error for different quantization levels.

Quantization	Asymptotic error
No quantization	5.30×10^{-8}
$\delta = 10^{-10}$	5.30×10^{-8}
$\delta = 10^{-8}$	7.36×10^{-8}
$\delta = 10^{-6}$	4.74×10^{-6}
$\delta = 10^{-4}$	5.64×10^{-4}
$\delta = 10^{-2}$	5.32×10^{-2}
$\delta = 10^{-1}$	4.91×10^{-1}

Asynchronous updates

Each agent completes the local update using $\theta = 10^{-8}$ with some probability $p \in \{\underline{p}, \bar{p}\}$ with $\underline{p} < \bar{p}$. The agents characterized by the smaller probability $\underline{p}$ are the "slow" nodes, which, having fewer computational resources, take on average a longer time to reach the threshold θ. Let N_s be the number of slow nodes.

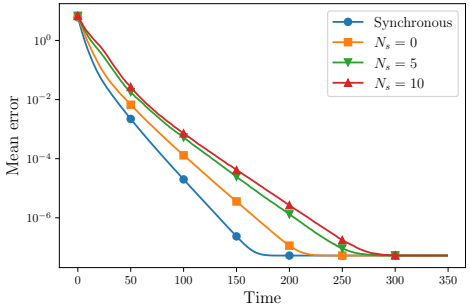


Figure: Error of DOT-ADMM with synchronous and asynchronous updates for different numbers of slow nodes.

Comparison with state-of-the-art algorithms

We compare DOT-ADMM with three gradient-based methods, for both static and online problems:

- *ra-GD*: gradient tracking algorithm which makes use of the robust ratio consensus to ensure convergence in the presence of asynchrony.

  Bof, N., Carli, R., Notarstefano, G., Schenato, L., & Varagnolo, D. (2018). Multiagent Newton–Raphson optimization over lossy networks. IEEE Transactions on Automatic Control, 64(7), 2983-2990.

- *LEAD*: gradient tracking algorithm which is designed to be robust to a certain class of unbiased quantizers.

  Liu, X., Li, Y., Wang, R., Tang, J., & Yan, M. (2020). Linear convergent decentralized optimization with compression. arXiv preprint arXiv:2007.00232.

- *DGD*: which does not converge exactly, but has been shown to be robust to additive errors and online scenarios.

  Yuan, K., Ling, Q., & Yin, W. (2016). On the convergence of decentralized gradient descent. SIAM Journal on Optimization, 26(3), 1835-1854.

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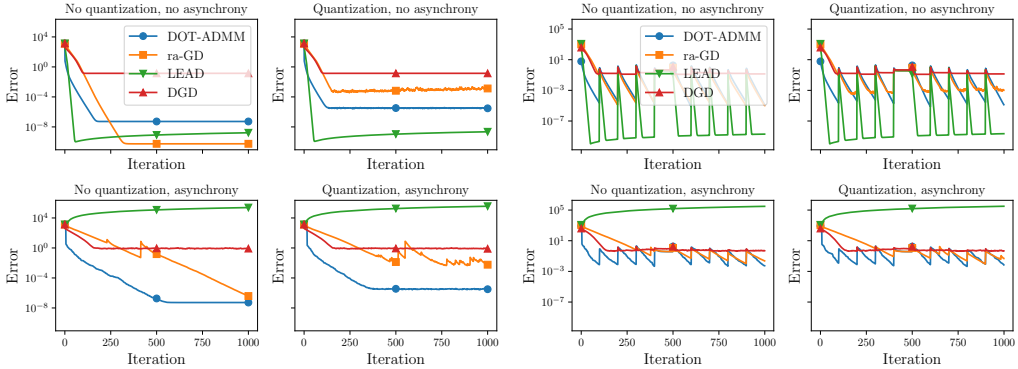


Figure: Comparison on static (left) and online (right) problems of DOT-ADMM with ra-GD, LEAD, DGD in different scenarios combining quantization/asynchrony.

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This work:

- N. Bastianello and D. Deplano, M. Franceschelli, K.H. Johansson, "Robust online learning over networks", IEEE Transactions on Automatic Control, 2025.

Application to dynamic consensus and personalized learning:

- D. Deplano, N. Bastianello, M. Franceschelli, K.H. Johansson, "A unified approach to solve the dynamic consensus on the average, maximum, and median values with linear convergence", IEEE Conference on Decision and Control, 2023.
- D. Deplano, N. Bastianello, M. Franceschelli, K.H. Johansson, "Distributed variance consensus with application to personalized learning", IEEE Conference on Decision and Control, 2023.

Ongoing extension to Open MAS:

- D. Deplano, N. Bastianello, M. Franceschelli, K.H. Johansson, "Optimization and learning in open multi-agent systems", IEEE Transactions on Automatic Control, (under review).

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Thank you for your attention!

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